

Coins and dice

1. Imagine that you will flip a coin eight times. Flipping eight times is your chance experiment.
 - a. What collection Ω of equally likely outcomes will you use as your sample space for this chance experiment?
 - b. How many outcomes are in the sample space Ω ?
 - c. How many outcomes are there in the event “flip exactly three heads”?
 - d. What is the probability of flipping exactly three heads?

2. Imagine you roll a standard six-sided die five times. Assume that the die is fair. Rolling a die five times is your chance experiment.
- a. What collection of equally likely outcomes will you use as your sample space for this chance experiment?
 - b. How many outcomes are in the sample space Ω ?
 - c. How many outcomes are there in the event “roll exactly two sixes”?
 - d. What is the probability of rolling exactly two sixes?